

APPLIED MATH FOR COMPUTER SCIENCE

CPCS 212

Lab 7: Solving linear systems

Gaussian Elimination

Open the diary file
Lab7_yourname.txt

Exercise 1:

$$x_1 + 2x_2 + 3x_3 = 6$$

$$2x_1 - 3x_2 + 2x_3 = 14$$

$$3x_1 + x_2 - 5x_3 = -2$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & -3 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \\ -2 \end{bmatrix}$$

```
>> % Exercise 1:
```

1-(a)

Create the coefficient matrix **A**, and the vector **b** for the above system.

```
>> % 1-(a)
>> A = [ 1  2  3; 2 -3  2; 3  1 -1 ]
>> b = [ 6; 14; -2 ]
```

1-(b)

Matlab provides a special operator, the backslash ($\backslash$) operator that is designed to solve a linear system.

```
>> % 1-(b)
```

```
>> x = A\b
```

```
x =
```

```
1.0000
```

```
-2.0000
```

```
3.0000
```

1-(c)

- Verify the answer by typing:

```
>> % 1 -( c)
```

```
>> A*x
```

Does $\mathbf{Ax} = \mathbf{b}$?

```
>> % ????????
```

1-(d)

Find the solution for the following systems using the
backslash (\) operator:

System 1:

$$x_1 + x_2 + x_3 = 4$$

$$2x_1 + 3x_2 + 4x_3 = 16$$

$$-2x_1 \quad -3x_3 = 11$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ -2 & 0 & -3 \end{bmatrix} \begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} = \begin{bmatrix} 4 \\ 16 \\ 11 \end{bmatrix}$$

```
>>% 1-(d) – System 1
```

```
>> A = ?????
```

```
>>b= ?????
```

```
>> x = ?????
```

System 2:

$$4x_1 + 2x_2 + 3x_3 + x_4 = 26$$

$$x_1 + 4x_2 + x_3 + 2x_4 + 5x_5 = 58$$

$$-2x_1 + 3x_2 + 5x_4 + 6x_5 = 73$$

$$x_1 + 4x_2 + 5x_3 + 2x_4 + 7x_5 = 72$$

$$x_2 + 3x_3 - 2x_4 + 4x_5 = 13$$

```
>>% 1-(d) – System 2
```

```
>> A = ?????
```

```
>>b= ?????
```

```
>> x = ?????
```

Exercise 2

- **Gaussian Elimination:**
- **(2 steps) :**
 1. **Forward Elimination.**
 2. **Back Substitution .**

Gaussian Forward Elimination:

form $\tilde{A} = [A \ b]$

for $i = 1 \dots n - 1$

for $k = i + 1 \dots n$

for $j = i \dots n + 1$

$$\tilde{a}_{kj} = \tilde{a}_{kj} - (\tilde{a}_{ki} / \tilde{a}_{ii}) \tilde{a}_{ij}$$

end

end

end

Gaussian Back Substitution :

given U, b

$$x_n = b_n / u_{nn}$$

for $i = n - 1 \dots 1$

$$s = b_i$$

for $j = i + 1 \dots n$

$$s = s - u_{i,j} x_j$$

end

$$x_i = s / u_{i,i}$$

end

$$x_n = \frac{b_n^{(n-1)}}{a_{nn}^{(n-1)}}$$

$$x_i = \frac{b_i^{(i-1)} - \sum_{j=i+1}^n a_{ij}^{(i-1)} x_j}{a_{ii}^{(i-1)}} \text{ for } i = n-1, \dots, 1$$

2-(a)

Create the augmented matrix (Ab) for the following system:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & -3 & 2 \\ 3 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \\ -2 \end{bmatrix}$$

2-(b) – First step - forward elimination

```
% --- Elimination

for i = 1:n-1      % loop over pivot rows
    pivot = Ab(i,i); % Ab = [A b];
    if abs(pivot)<ptol
        error('zero pivot encountered');
    end
    for k = i+1:n    % k = index of next row to be eliminated
        Ab(k,i:nb) = Ab(k,i:nb) - (Ab(k,i)/pivot)*Ab(i,i:nb);
                                                % nb = n+1;
    end
    fprintf('\nAfter elimination in column %d with pivot = %f\n',i,pivot);
    disp(Ab);
end
```

2-(c)

There is a variable called *ptol*.

What it is used for?

```
>>% 2- (c)
```

```
>> ptol = a pre-specified error tolerance,
```

2-(d) Second step -back substitution

```
% --- Back substitution

% preallocate memory for and initialize x
x = zeros(n,1);
x(n) = Ab(n,nb)/Ab(n,n);
for i=n-1:-1:1
    x(i) = (Ab(i,nb) - Ab(i,i+1:n)*x(i+1:n))/Ab(i,i);
end
```

2-(e)

Use the codes given above as a model for an m-file called ***GE_noPiv.m*** that performs Gaussian elimination without pivoting. The function should start out with:

GE_noPiv.m

```
function x = GE_noPiv(A,b,ptol)
% GE_noPiv Show steps in Gauss elimination and back
% substitution
% No pivoting used.
%
% x = GE_noPiv(A,b,ptol)
% x = GE_noPiv(A,b)
%
% Input : A,b = ????   Coefficient Matrix,
%      ptol = (optional)???? a pre-specified error tolerance,
%      default:   ptol = 50 * eps
%
% Output:   x = solution vector, if solution exists   Output Matrix
```

GE_noPiv.m

```
function x = GE_noPiv(A,b,ptol)
% comments
if nargin<3
    ptol = 50*eps;
end
Ab = [A b];
n = length(A);
nb = n+1;
% --- Elimination
% --- Back substitution
```

```
function x = GE_noPiv(A,b,ptol)
```

```
% comments
```

```
if nargin<3
```

```
ptol=50*eps;
```

```
end
```

```
Ab = [A b];
```

```
n= length(A);
```

```
% or n=max(size(A));
```

```
nb = n+1;
```

```
for i = 1:n-1
```

```
    pivot = Ab(i,i);
```

```
    if abs(pivot)<ptol
```

```
        error('zero pivot encountered');
```

```
    end
```

```
% --- Elimination
```

```
for k = i+1:n
```

```
Ab(k,i:nb) = Ab(k,i:nb) - (Ab(k,i)/pivot)*Ab(i,i:nb);
```

```
end
```

```
fprintf('\nAfter elimination in column %d with pivot = %f\n',i,pivot);
```

```
disp(Ab);
```

```
end
```

```
% --- Back substitution
```

```
x = zeros(n,1);
```

```
x(n) = Ab(n,nb)/Ab(n,n);
```

```
for i=n-1:-1:1
```

```
x(i) = (Ab(i,nb) - Ab(i,i+1:n)*x(i+1:n))/Ab(i,i);
```

```
end
```

```
end % end of function
```

2-(f)

Test your *GE_noPiv* function with the systems in exercise 1-(d). Are the solutions the same with the obtained using the backslash operator?

2-(f) System 1:

$$x_1 + x_2 + x_3 = 4$$

$$2x_1 + 3x_2 + 4x_3 = 16$$

$$-2x_1 \quad -3x_3 = 11$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ -2 & 0 & -3 \end{bmatrix} \begin{bmatrix} x1 \\ x2 \\ x3 \end{bmatrix} = \begin{bmatrix} 4 \\ 16 \\ 11 \end{bmatrix}$$

```
>>% 2-(f) – System 1
```

```
>> A = ?????
```

```
>> b = ?????
```

```
>> x = GE_noPiv( A,b)
```

2-(f) System 2:

$$4x_1 + 2x_2 + 3x_3 + x_4 = 26$$

$$x_1 + 4x_2 + x_3 + 2x_4 + 5x_5 = 58$$

$$-2x_1 + 3x_2 + 5x_4 + 6x_5 = 73$$

$$x_1 + 4x_2 + 5x_3 + 2x_4 + 7x_5 = 72$$

$$x_2 + 3x_3 - 2x_4 + 4x_5 = 13$$

```
>>% 2-(f) – System 2
```

```
>> A = ????
```

```
>> b = ????
```

```
>> x = GE_noPiv( A,b,eps)
```

2-(g) - HomeWork

- Test your ***GE_noPiv*** function with the following systems. Verify that your code can handles these situations.
- Find the solution for the following systems using the backslash (\) operator. Verify that (\) can handles these situations.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & -2 & 5 \\ 5 & -1 & 7 \\ 9 & 7 & 6 \\ -8 & 3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \\ 6 \\ 17 \\ 45 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & -2 & -2 \\ 1 & 2 & 4 & -3 \\ -3 & -3 & 8 & -2 \\ -1 & 1 & 6 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -4 \\ 5 \\ 7 \\ 7 \end{bmatrix}$$